

Priyanshu Verma

System Engineer | Spring Boot · Apache Kafka · Microservices | FinTech & Banking Systems

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PROFESSIONAL SUMMARY

Java Backend Engineer with 4 years designing high-availability distributed systems for Tier-1 banks. Onsite Technical Lead for **TCS BaNCS** leading **Corporate Actions** at Rand Merchant Bank, ABSA, and Standard Bank (Johannesburg), lead the platform's **first Cloud deployment**. Deep expertise in **Spring Boot, Apache Kafka, Microservices, Redis, PostgreSQL**, and event-driven architecture. Maintained 99.95% uptime on SWIFT / ISO 20022-compliant payment systems processing real money at scale. 6x award winner for technical excellence.

TECHNICAL SKILLS

Languages: Java (JVM, Java 21), Python, SQL, XML, HTML/CSS

Backend: Spring Boot, Spring Cloud, Spring Security, Spring Batch, Microservices, Apache Kafka, Hibernate/JPA, REST APIs

Cloud & DevOps: Azure, AWS (EC2, S3, RDS, SQS), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Linux

Data & Caching: PostgreSQL, Oracle DB, Cassandra, Redis, pgvector

AI / ML: Azure OpenAI (GPT-4o), RAG

Reliability & Testing: Resilience4j, JUnit 5, Mockito, Testcontainers, Postman

Banking Domain: TCS BaNCS, SWIFT, ISO 20022, DTCC, Payment Systems, Reconciliation | Agile, Scrum, JIRA, Git, Maven

PROFESSIONAL EXPERIENCE

System Engineer — Onsite Technical Lead | TCS BaNCS

Aug 2022 – Present

Tata Consultancy Services | Johannesburg, South Africa

- Led critical core-banking **Corporate Actions** initiatives at **Rand Merchant Bank, ABSA, and Standard Bank** as the single technical point of contact for business, product, and architecture stakeholders across Tier-1 banks.
- Architected and developed **event-driven Kafka-based microservices** for an Electronic Voting System and Email Notification Platform, processing real-time **Corporate Actions** (eligibility, entitlements, tax, reconciliation) while delivering instant notifications to 50,000+ customers with zero-downtime deployments and horizontally scalable, schema-evolving architecture.
- Developed end to end high-availability **Tax Management Interfaces and Batches** (Java, Spring Framework, Oracle DB) achieving 99.9% transaction accuracy, 40% faster reconciliation, and 35% lower API response times.
- Built event-driven microservices with **Apache Kafka, Spring Batch, and AWS SQS**; delivered AI-powered notifications via **Azure OpenAI** serving 50,000+ customers and improving workflow efficiency by 60%.
- Raised system reliability from 85% to 98% via **Jenkins CI/CD** automation; reduced production incidents by 45% while sustaining 99.95% uptime on **SWIFT, ISO 20022, and DTCC-compliant** systems.

Analyst

Dec 2021 – Jul 2022

Capgemini | Mumbai, India

- Developed Bank Management Systems (**Java, Spring Boot, PostgreSQL, MongoDB**); containerized deployments with **Docker & Kubernetes on AWS EC2/S3** — reducing deployment time by 65%.
- Built **RESTful APIs** for distributed communication; drove code reviews, performance tuning, and test automation with JUnit, Mockito, and Testcontainers in Agile sprints.

PROJECTS

SentinelFlow — AI-Powered Fraud Detection Platform

[GitHub](#) · [Live Portfolio](#)

- Architected **6 Spring Boot microservices** for real-time fraud scoring over an event-driven **Kafka pipeline with exactly-once semantics** (Outbox pattern, idempotent producers) on PostgreSQL and Redis.
- Built a distributed **sliding-window rate limiter (Redis Lua scripts)** with Resilience4j circuit breakers, benchmarked at sub-20ms p99; automated quality gates via **GitHub Actions + Testcontainers**.
- Engineered **RAG-based fraud explanations (Azure OpenAI GPT-4o + pgvector)** with Redis semantic caching to cut LLM cost, streamed to a React UI over SSE; full observability with OpenTelemetry, Prometheus, and Grafana.

PayFlow — Event-Driven Payment System

[GitHub](#)

- Built a **Kubernetes-deployed** payment platform (Spring Boot, Kafka, Redis, PostgreSQL); diagnosed and resolved real distributed-systems failures — Kafka networking, Redis serialization, and image-pull issues — documented end to end.

EDUCATION

B.Tech — Electronics & Communication Engineering | Pranveer Singh Institute of Technology, Kanpur | CGPA: 9.0 / 10 | 2018 – 2022

AWARDS & RECOGNITION

- Employee of the Quarter** — Exceptional technical performance & on-time project delivery at TCS
- Star of the Month (2x)** — Outstanding contributions in development, design & system architecture
- On the Spot Award (2x)** — Rapid resolution of production-critical P1/P2 incidents under tight SLAs
- Best Team Award** — Collaborative success across complex multi-bank BaNCS implementations